

CAM/AUTO: Board Submission

Supervised production in 6 months, with the first learning module | 29 September 2026

1. The ask

Approve **INR 45 to 65 lakh** and a team of **2 to 3 developers**, the founder (half time) and an in-house process owner (half time) to take CAM/AUTO live for standard 3-axis milled parts, **26 weeks after the geometry lead joins** (about 8 to 9 months from today), and to ship the **first version of its learning module** in the same period. The learning module is the reason to take this app further: it is what lets every part we cut make the next program better.

- **Outcome:** the shop programs new prismatic jobs in CAM/AUTO instead of the commercial CAM seat, with the process owner approving every program.
- **Proof:** 30 real parts machined and measured, 27 or more right on the first cut, zero gouges (15 parts, 13 or more, at the minimum budget).
- **Learning:** from week 3 every cut is recorded; by week 21 the app proposes its own cutting settings and size corrections from that data, each one approved by the process owner before use.
- **Money:** INR 45 to 65 lakh with three developers, INR 32 to 45 lakh with two; a stop-or-continue review at week 11.
- **Limit:** supervised production only. Curved surfaces, lathe, 4-axis and unattended release are phase two.

2. Why accuracy stopped improving, and what changes

After 79 releases and two shop trials, the app works end to end but its engine has five built-in limits. Everything else (screens, drawing reader, planning rules, operator instructions, tests) stays; the engine is replaced from the inside, behind the same function names, without a rewrite.

Limit today	What the shop saw	Fix in this plan
Reads the 3D model as a rough 0.4 mm grid, not the exact shape	Walls 0.3 mm oversize, wavy edges, holes placed to grid accuracy	Read the exact CAD model (full OpenCascade kernel)
Cutting paths follow the grid	Stepped, oversize profiles (already fixed for the outer profile in R78)	Exact paths with true arcs on every wall
The safety check uses the same data that wrote the path	Errors pass as clean	A second, independent check of the final machine program
Nothing comes back from the machine	Defects arrive as photos weeks later; nothing can learn	Program ID, measurement form and a shop database from week 3: the data the learning module runs on
One 12,600-line file, no automatic checks	Every fix needs a regression hunt	Split into modules with automatic checks, by Claude Code, before the hires join

3. The learning module: what it learns, what it never touches

Learning is a data loop with guarded models, not a model that draws toolpaths. Geometry stays exact and deterministic; learning tunes the decisions around it. Version 1 ships in this plan; it grows in phase two as data accumulates.

Learns in 6 months (v1)	Learns in phase two	Never learns
Cutting speeds and feeds per material, tool and machine, from measured results and cycle times	Which strategy and tool to pick for each feature type, from first-article results	The shape of the part, the toolpath geometry, offsets and depths
Size corrections per machine and tool, from CMM deviations	Setup and clamping choices, from setup time and rework	The safety checks and collision rules
Drawing-reading corrections, from what operators fix on screen	Wider models once 100+ measured parts exist	The machine program output and datum logic

Guardrails the board should insist on: every learned value is replayed on the whole part library before release; the process owner approves it; nothing that fails the safety check ever ships; and the one number that judges it is **first-article corrections per part, quarter over quarter**.

4. Team

Role	Who	Time	Owns
Geometry lead	New hire, 6 to 10 years CAM or computational geometry	Full time from wk 1	Exact geometry, independent check, code review
Platform and test engineer	New hire, 4 to 7 years	Full time from wk 1	Modules, checks, part library, measurement loop, learning data pipeline
Application developer	New hire or contractor, 3 to 6 years (3-developer scope)	From wk 4	Shop tablet page, database, learning v1 models, Siemens post
Founder	Internal	Half time	Product decisions, Claude Code sessions, weekly gate review
Process owner	Internal senior CNC programmer; else a fourth hire	Half time from approval	Approves every program and learned value; runs and measures first articles
Claude Code	AI coding tool on every desk	Continuous	Refactors, tests, posts, screens, first drafts; always behind checks and human review

5. Plan and gates

Weeks count from the geometry lead's joining date, expected 8 to 12 weeks after approval because senior engineers serve 60 to 90 day notice periods. The weeks before that are not idle: the founder, the process owner and Claude Code prepare the code and the measurement loop.

Weeks	Phase	Gate the board can see
Before wk 1	Preparation (8 to 12 weeks while hiring): code split into modules, automatic checks, measurement form, program ID, today's accuracy recorded as the baseline	Lead's joining date fixed
1 to 3	Foundation: hires onboarded, test library of parts, shop database live	Checks green, baseline measured
3 to 11	Exact geometry: exact model reading, exact paths on every wall	Paths correct to 0.01 mm on 15 parts. Stop-or-continue point
8 to 15	Second check: independent re-check of every program; dry runs on the shop machines	Second check live, posts clean
12 to 21	Shop loop and learning v1: weekly first articles with CMM records; learned speeds, feeds and size corrections proposed from the data and approved before use	27 of 30 parts right first time (13 of 15 at minimum); learning v1 in use
20 to 26	Go-live: hardening, operator training, three jobs run by setters from the instructions alone	Sign-off by process owner and founder

6. Cost, risks and decisions

Item (6 months)	Minimum, 2 developers	Full, 3 developers
Developers	29 to 39 lakh	37 to 51 lakh
Claude Code seats and usage	1.5 to 2 lakh	1.5 to 3 lakh
Workstations, on-site server, backup	2.5 to 4.5 lakh	4.5 to 7 lakh
Machine and CMM time for first articles (internal rates)	2 to 3 lakh	3 to 5 lakh
Recruiting and 10 percent contingency	5 to 7 lakh	7 to 10 lakh
Total	32 to 45 lakh	45 to 65 lakh
If no in-house process owner can be freed: fourth hire, outside the total	+7.5 to 10 lakh	+7.5 to 10 lakh

Salary bands are Mumbai 2026 estimates to validate with a recruiter. With two developers, learning v1 is limited to size corrections and data collection; speeds-and-feeds learning and the tablet page need the third developer. This plan is phase one of the 12-month Board Brief: cheaper and faster because the scope is narrower, not because steps are skipped.

Main risks. The geometry lead joining later than 12 weeks after approval moves go-live one for one (retained search starts now). The shop not recording measurements weekly removes both the proof and the learning data (no record, no sign-off). Claude Code output accepted without review can ship subtle geometry errors (nothing merges without checks and a human reading the change).

Cut list if week 11 is late, in order: Siemens post validation; faster roughing; on-site tablet page; part library from 30 to 15. **Never cut:** the go-live date, zero gouges, process-owner approval of every program, and the learning data loop.

Decisions asked today:

1. Approve the 6-month plan at INR 45 to 65 lakh (full) or INR 32 to 45 lakh (minimum), with a stop-or-continue review at week 11.
2. Confirm the go-live definition: supervised 3-axis production with process-owner approval on every program, and learning v1 in use under the same approval.
3. Name the process owner (half time from approval), or approve the fourth hire.
4. Approve recruiting the geometry lead and the platform engineer now, retained search on the lead.
5. Reserve weekly machine and CMM hours for first articles from week 3.
6. Accept that the 26-week clock starts on the geometry lead's joining date.